Write Your Project Title Here (Clear, Specific, and Technical)

*Team ID: Write Your Assigned Team ID Here

Student Name 1 (Roll No: XXXXXXXX)
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***Abstract*—The abstract must concisely summarize the entire project in approximately 150 to 250 words. It should begin by clearly stating the problem being addressed and explaining its practical importance. The abstract should then briefly describe the proposed methodology, including the type of model or algorithm used. Students must also mention the dataset employed for experimentation and summarize the most important quantitative results using appropriate evaluation metrics such as accuracy, F1-score, PSNR, SSIM, or other task-specific measures. The abstract should conclude with a short statement highlighting the overall contribution of the work. Citations should not be included in the abstract, and background explanations should be avoided.**

***Index Terms*—Students must provide four to six technical keywords that accurately describe the core concepts of the project. Keywords should be separated by commas and must reflect the main techniques, application domain, and methodologies used.**

I. Introduction

The introduction section should provide a comprehensive overview of the problem addressed in the project. Students must clearly define the problem statement and explain its relevance in real-world scenarios. This section should describe the broader context in which the problem exists and discuss why solving it is important from a societal, industrial, or technological perspective.

After introducing the problem, students should briefly discuss existing approaches and identify their limitations. The introduction must not become a detailed literature survey, but it should highlight the general shortcomings of current solutions. Following this discussion, students should introduce their proposed approach and explain how it aims to address the identified limitations.

The introduction should end with a clear statement of contributions. These contributions must be specific and measurable, such as proposing a new architecture, improving performance metrics, introducing a novel preprocessing technique, or conducting a comprehensive comparative analysis. The length of this section should typically be around one page.

II. Literature Survey

The literature survey must demonstrate that students have reviewed relevant academic research related to their project topic. A minimum of six to ten research papers should be included, preferably published between 2019 and 2026. Each referenced study should be summarized in the students' own words.

For every paper discussed, students should explain the problem addressed by the authors, the methodology used, the dataset involved (if applicable), and the key findings. It is also essential to mention the limitations or gaps identified in each study. This analysis will help establish the research gap that motivates the current project.

The section should conclude with a synthesis paragraph that clearly explains what is missing in existing literature and how the proposed project intends to address those shortcomings. Proper IEEE citation style must be followed throughout the section.

III. Proposed Methodology

The proposed methodology section is the most critical part of the report. It should provide a detailed explanation of the system design and implementation so that another researcher could replicate the work.

Students must begin by describing the overall architecture of the system. A block diagram or workflow figure should be included to visually represent the processing pipeline. The section should clearly explain the input data, preprocessing steps, feature extraction techniques (if applicable), and the machine learning or deep learning model used.

If a mathematical formulation is involved, all equations must be clearly defined and properly formatted. For example, if classification accuracy is used, it may be defined as:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{1}$$

All symbols used in equations must be explained in the text. The loss function, optimization strategy, and training procedure should also be described in detail. Students should specify how the model makes predictions during the inference stage.

IV. Experimental Results

The experimental results section must present quantitative evidence supporting the effectiveness of the proposed approach. Students should begin by describing the computational environment, including hardware specifications such as CPU, GPU, and RAM, as well as software frameworks used for implementation.

The dataset split ratio (for example, 80% training and 20% testing) must be clearly stated. Important training parameters such as number of epochs, batch size, learning rate, and optimizer should be reported.

Results should be presented in well-formatted tables and figures rather than screenshots. Performance metrics must be analyzed and interpreted. Students should explain why certain results were achieved, whether overfitting or underfitting occurred, and how their model compares to baseline approaches.

A. Dataset Description

In this subsection, students must describe the dataset in detail. The dataset name, source, number of samples, number of classes, and distribution characteristics should be clearly specified. If the dataset is imbalanced, this should be mentioned. For image-based projects, representative sample images may be included.

If a custom dataset is used, students must explain how the data was collected, any preprocessing performed, and any ethical considerations involved in data acquisition.

B. Evaluation Metrics

All evaluation metrics used in the project must be clearly defined and explained. For classification tasks, students should define accuracy, precision, recall, and F1-score. Mathematical expressions must be written in proper format. Each metric should be explained conceptually so that the reader understands what aspect of performance it measures.

C. Comparison with State-of-the-Art Methods

In this subsection, students are required to compare their proposed method with at least two or three baseline or state-of-the-art methods. The comparison must be quantitative and based on consistent evaluation metrics across all methods. Students should ensure that the comparison is fair, meaning that all models are evaluated under the same dataset split and experimental conditions whenever possible.

The results must be presented in a structured tabular format. The table should include the name of each method and the corresponding performance metrics. An example comparison table is shown below for guidance.

After presenting the comparison table, students must provide a detailed analysis of the results. The discussion should clearly explain whether the proposed method outperforms existing approaches and identify the specific metrics where improvements are observed. If the proposed method performs similarly or worse in certain metrics, students must provide logical explanations, such as model complexity, dataset imbalance, or limited training data.

Students are expected to interpret the results rather than merely report numerical values. A fair, transparent, and academically honest comparison is required.

TABLE I
Performance comparison with baseline and state-of-the-art methods.

Method	Accuracy	Precision	Recall	F1-score
Logistic Regression	0.84	0.82	0.80	0.81
Support Vector Machine	0.88	0.86	0.85	0.85
CNN (Baseline)	0.91	0.90	0.89	0.89
Proposed Method	0.94	0.93	0.92	0.92

TABLE II
Effect of learning rate on validation accuracy.

Learning Rate	Validation Accuracy
0.01	0.86
0.001	0.91
0.0001	0.94
0.00001	0.90

D. Hyperparameter Tuning

In this subsection, students must demonstrate how hyperparameters were systematically tuned rather than simply reporting the final selected values. Hyperparameter tuning refers to the experimental process of evaluating multiple parameter configurations and selecting the combination that yields the best validation performance.

Students should clearly describe the tuning strategy adopted, such as grid search, random search, or iterative experimental adjustment. The tuning process must involve testing multiple candidate values and evaluating performance on a validation dataset.

For example, the learning rate may be tested over several values as shown below.

From Table II, it can be observed that a learning rate of 0.0001 provided the highest validation accuracy. Larger learning rates resulted in unstable convergence, while excessively small learning rates slowed down training and reduced performance.

Similarly, other hyperparameters such as batch size and number of epochs should also be evaluated experimentally. An example of batch size tuning is shown below.

Based on the experimental observations, students must justify the final selected configuration. The final hyperparameter values should be reported only after presenting the tuning analysis. The justification must explain why certain values were chosen and why others were rejected, demonstrating un-

TABLE III
EFFECT OF BATCH SIZE ON VALIDATION PERFORMANCE.

Batch Size	Validation Accuracy
16	0.92
32	0.94
64	0.91

derstanding of model convergence behavior and generalization performance.

V. CONCLUSION

The conclusion section should summarize the entire project in a concise manner. Students should restate the problem addressed, briefly describe the proposed solution, and highlight the most significant experimental findings.

The conclusion should also acknowledge limitations of the current work, such as limited dataset size or computational constraints. Finally, students should suggest potential future research directions or possible improvements to the system.

No new experimental results should be introduced in this section.

REFERENCES

[1] Author Name, "Title of paper," *Journal Name*, vol. X, no. Y, pp. 00–00, Year.

[2] Author Name, *Book Title*. Publisher, Year.

REFERENCE WRITING INSTRUCTIONS

All references must follow IEEE format. A minimum of 15–20 scholarly references should be included. Students are advised to use Google Scholar and select the IEEE citation format when exporting references.

PAGE LIMIT AND FORMATTING REQUIREMENTS

This mini-project report must strictly follow the IEEE conference format provided in this template. The total length of the paper must be a minimum of six pages and should not exceed seven pages, including figures, tables, and references.

The recommended page-wise distribution is as follows:

- The Introduction section should occupy approximately one full page and must clearly define the problem statement, motivation, and contributions. The Literature Survey section should also be approximately one page and must include a critical review of relevant recent research papers.
- The Proposed Methodology section should span approximately one and a half pages. This section must provide a detailed explanation of the system architecture, algorithmic design, mathematical formulation (if applicable), and implementation workflow.
- The Experimental Results section should also span approximately one and a half pages. It must include dataset description, evaluation metrics, quantitative comparisons, tables, figures, and performance analysis.
- The Abstract should be limited to one well-structured paragraph of 150–250 words. The Conclusion should also be written in a single concise paragraph summarizing the findings, limitations, and future scope.
- The paper must include a minimum of fifteen to twenty scholarly references formatted strictly according to IEEE citation style. All listed references must be cited within the text.
- Submissions that do not meet the minimum page requirement or fail to follow the prescribed structure may not be evaluated.